Exercise: Scientific Calculator

Overview

In this project students will build a command-line scientific calculator. The program will display a menu of options including several arithmetic operations and options to clear the result, display statistics, and exit the program. The project's goal is to provide students with practice in looping, type conversion, and data persistence.

Specification

The program must have the **`main()`** function as the entry point, which must only run if invoked directly – i.e., by checking `__name__`. There should be ***no global variables***. All output should go to standard output unless otherwise specified. When the program starts it should display a menu, prompt the user to enter a menu option, and read a value:

```
Current Result: 0.0

Calculator Menu
---------------
0. Exit Program
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exponentiation
6. Logarithm
7. Display Average

Enter Menu Selection: 1
```

If an option with operands (1-6) is selected, the program should prompt for and read floating point numbers as follows:

```
Enter first operand: 89.1
Enter second operand: 42
```

Once the two operands have been read, the result should be calculated and displayed, along with the menu:

```
Current Result: 131.1

Calculator Menu
---------------
…
```

Operational Behavior

This calculator includes multiple behaviors that are unique depending on the input and operation specified; they are detailed in this section. Results should be displayed to default precision except where specified.

Exponentiation

For exponentiation, the first operand should be used as the base and the second as the exponent, i.e.:

If the first operand is 2 and the second is 4… $2^4 = 16$

Logarithm

For logarithms, the first operand should be used as the base and the second as the yield, i.e.:

If the first operand is 2 and the second is 4… $\log_2 4 = 2$

Displaying the Average

As the program progresses, it should store the total of all results of calculation and the number of calculations. Note that this does **not** include the starting value of 0! The program should display the average of all calculations as follows:

```
Sum of calculations: 101.3
Number of calculations: 2
Average of calculations: 50.15
```

Note that the average calculation should show a maximum of **two** decimal places. The program should immediately prompt the user for the next menu option (without redisplaying the menu).

If no calculations have been performed, this message should be displayed:

```
Error: No calculations yet to average!
```

Using Results of Calculation

The program should allow the user to use the previous result in an operation. The user may enter the word "**RESULT**" in place of an operand; if the user does so, the program should replace this operand with the result of the previous calculation (or zero if this is the first calculation):

```
Enter first operand: 89.1
Enter second operand: RESULT
```

Submissions

NOTE: Your output must match the example output *exactly*. If it does not, ***you will not receive full credit for your submission***!

File: SciCalculator.py
Method: Submit on Canvas

Do not submit any other files!